MLOps

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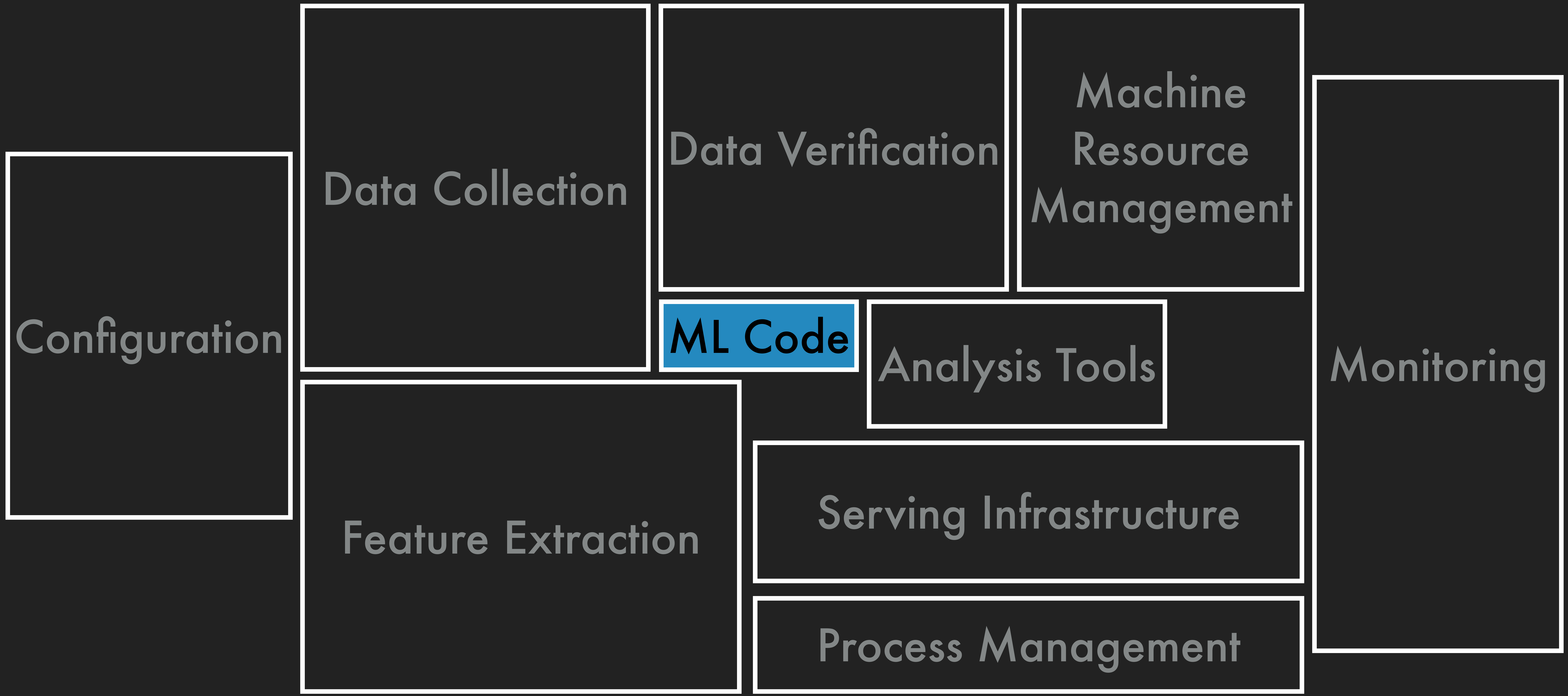
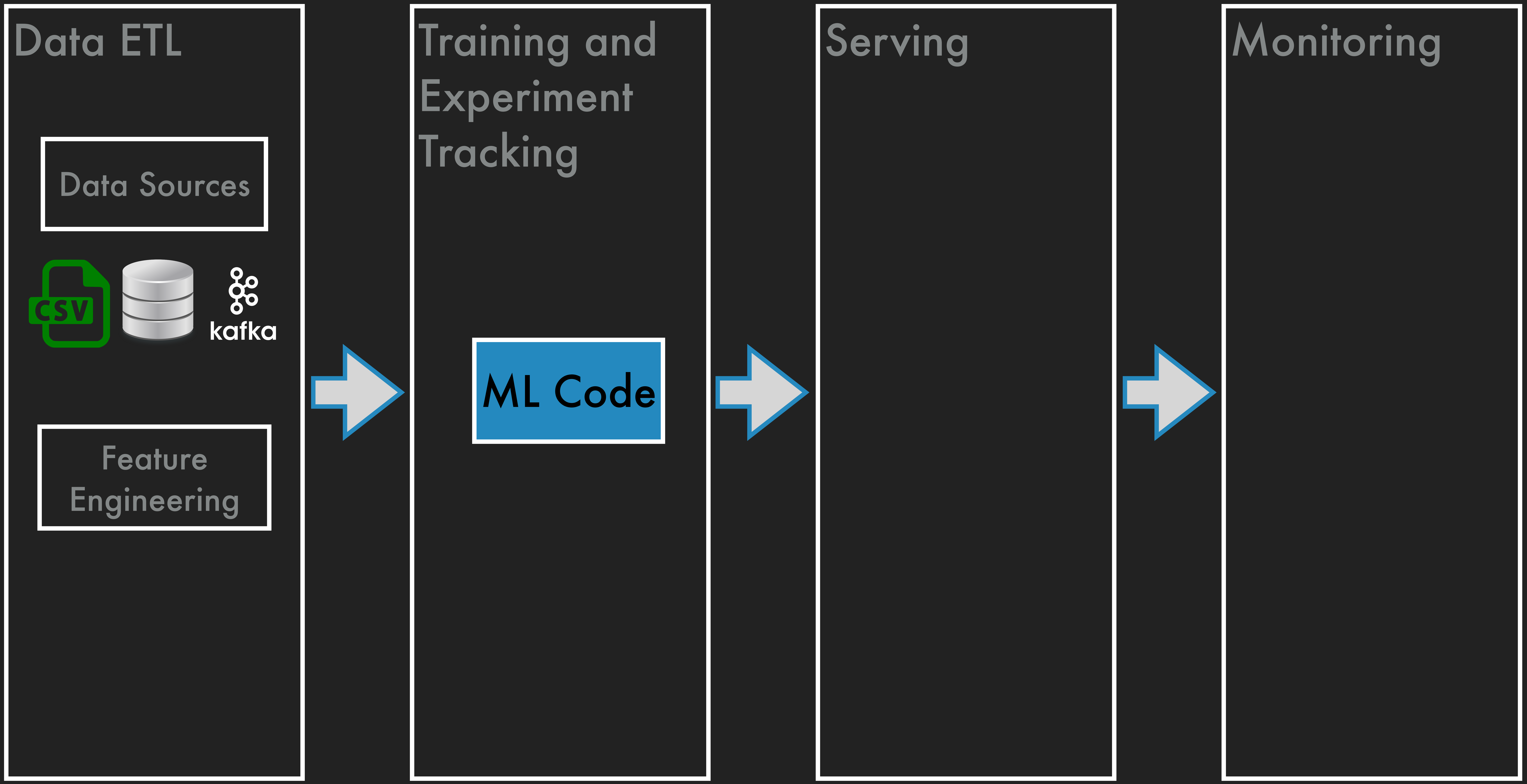
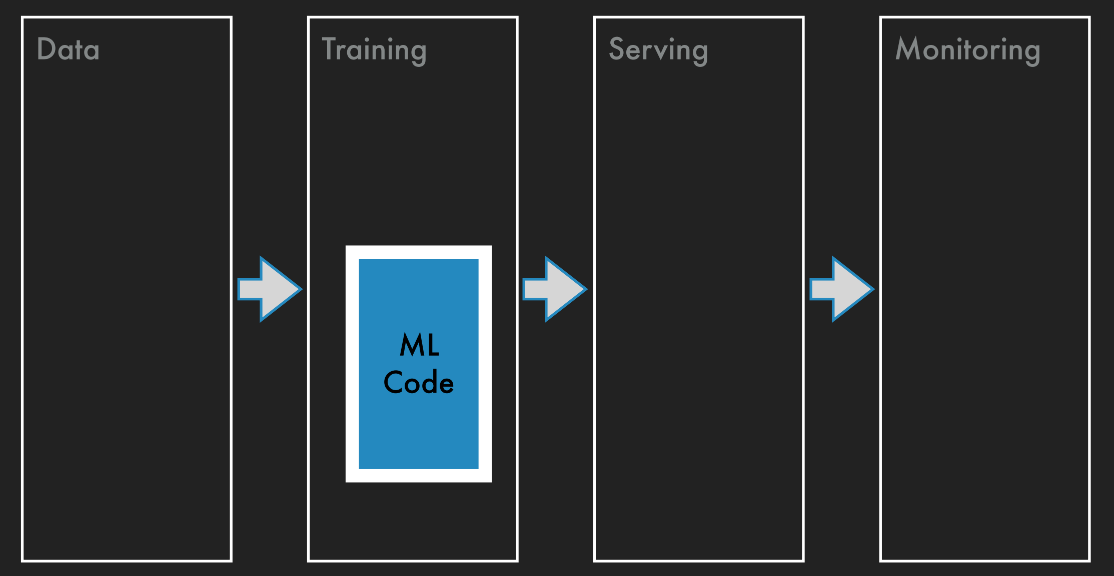
Operationalizing prototype ML models into scalable, collaborative, and automated production ML systems that solve complex real-world problems



Adapted from [Hidden Technical Debt in Machine Learning Systems]

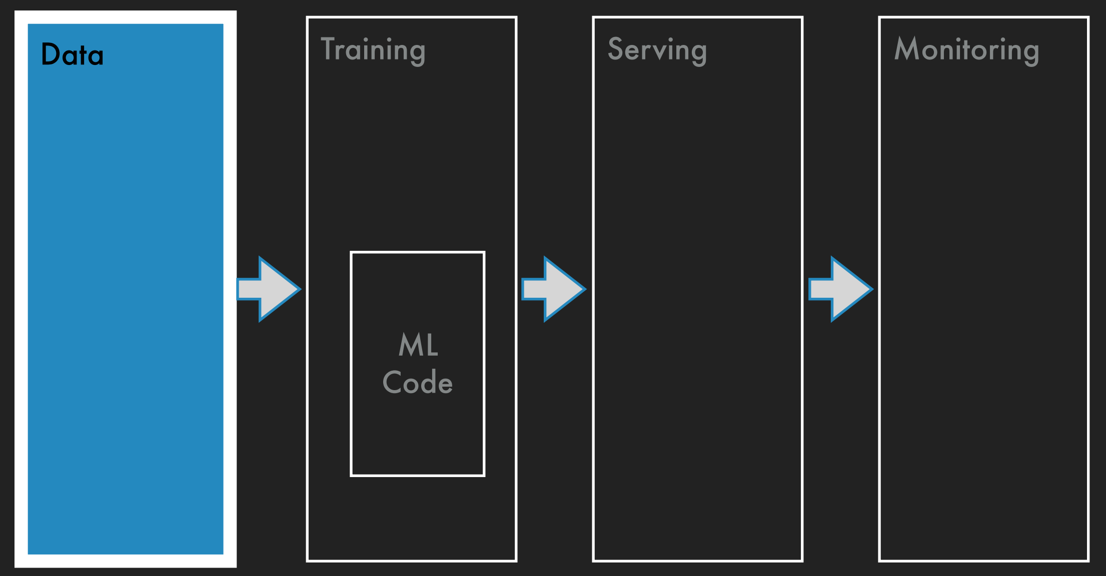
Predict if the credit card transaction is fraudulent


Data ETL
Data Sources
CSV
kafka
Feature Engineering
Training and Experiment Tracking
ML Code
Serving
Monitoring



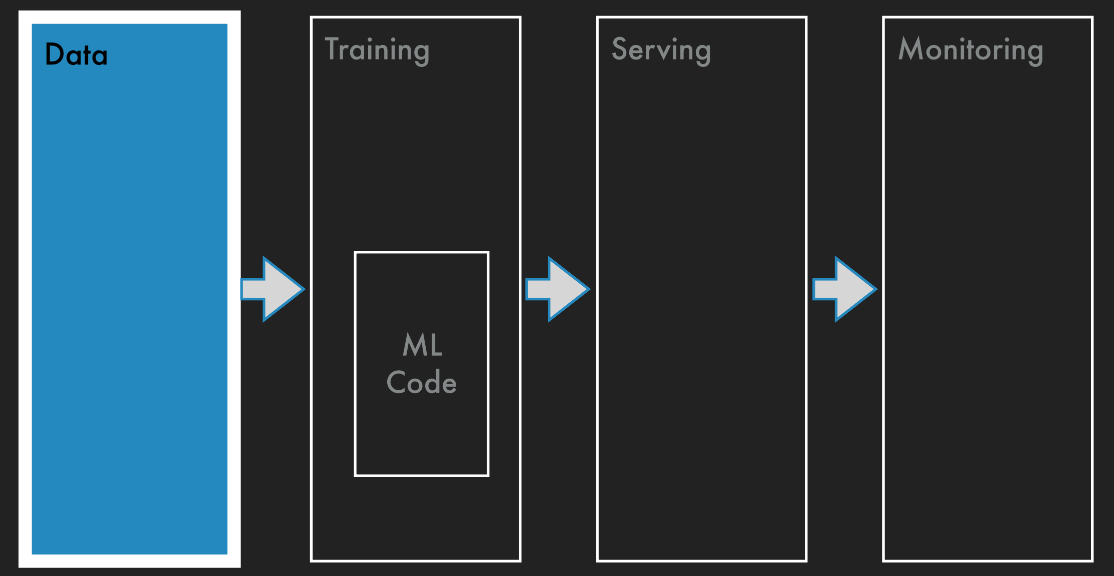
Which ML models could you use?

- Classification models
  - Logistic regression
  - SVC
  - Decision trees
  - XGBoost
  - Neural networks



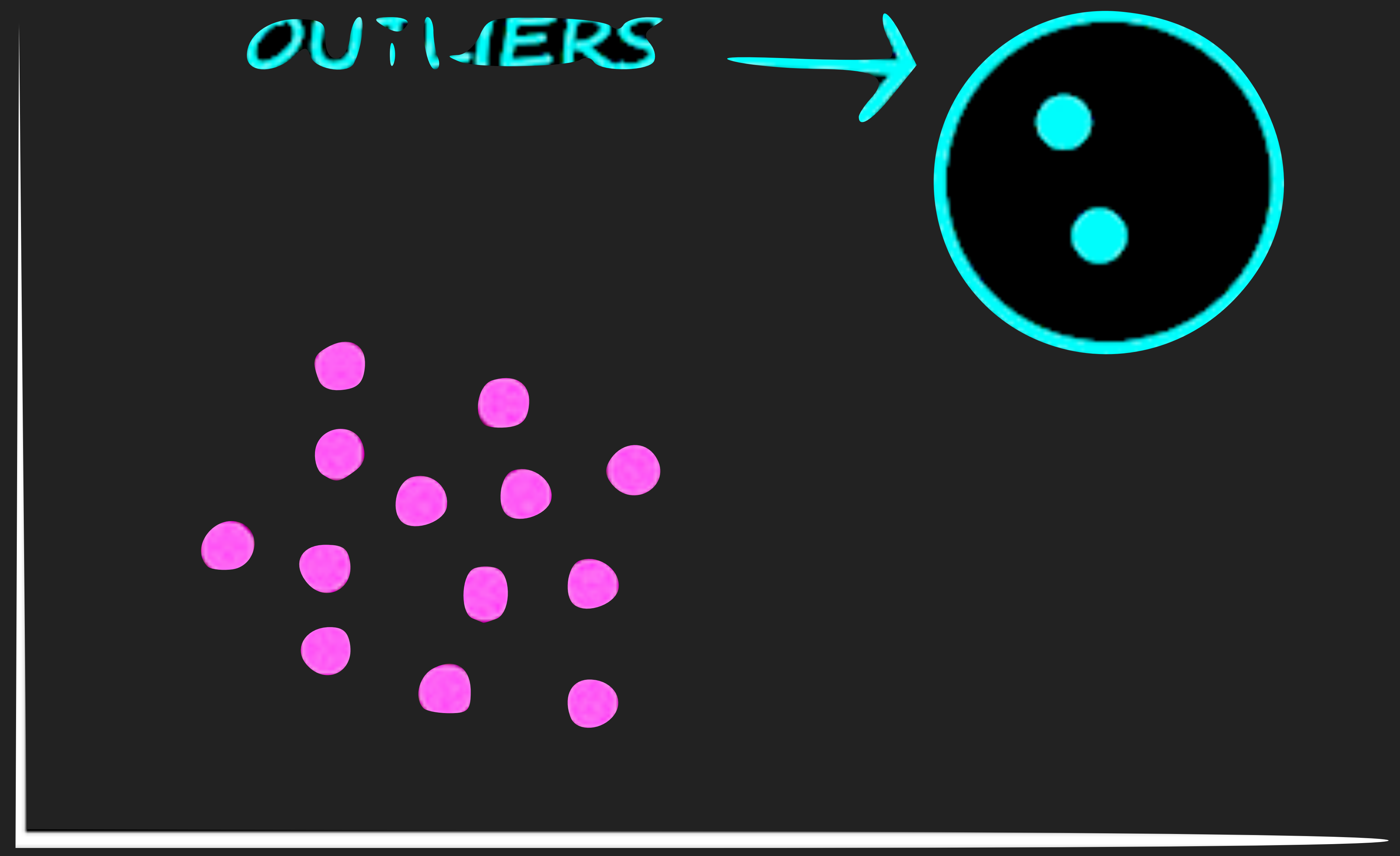
Data Collection

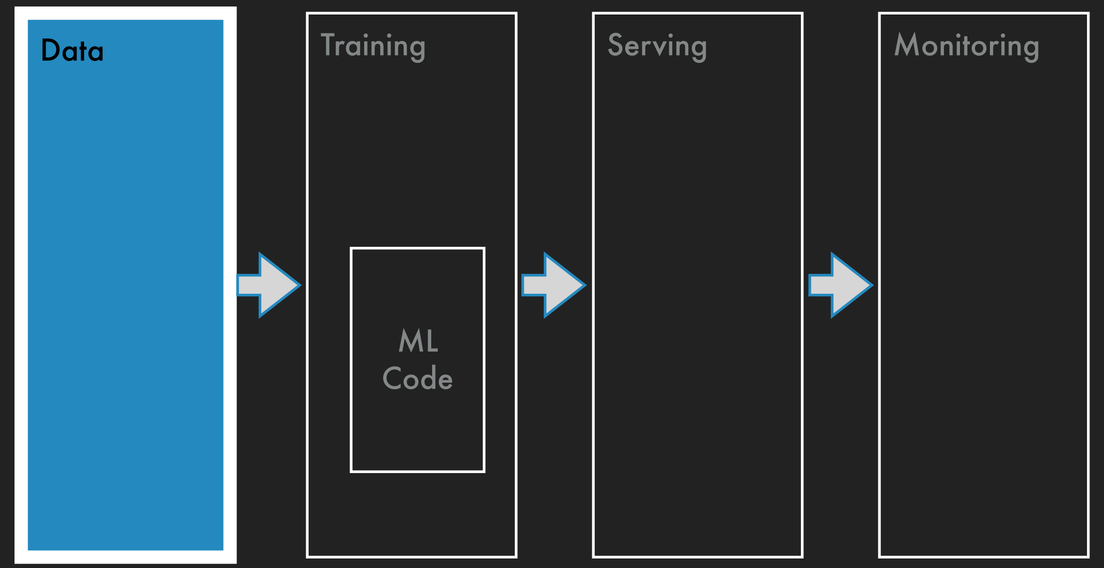
- Labels
  - Chargebacks - 0/1
  - Reason codes
  - Suspected fraud
- Features
  - Raw: Amount, hour of day, currency
  - Engineered: log(amount), distance between merchant and buyer, average amount spent in last 10 days
- Scale
  - 1M to 100M daily transactions
  - 1 – 2 years of data, 10B rows



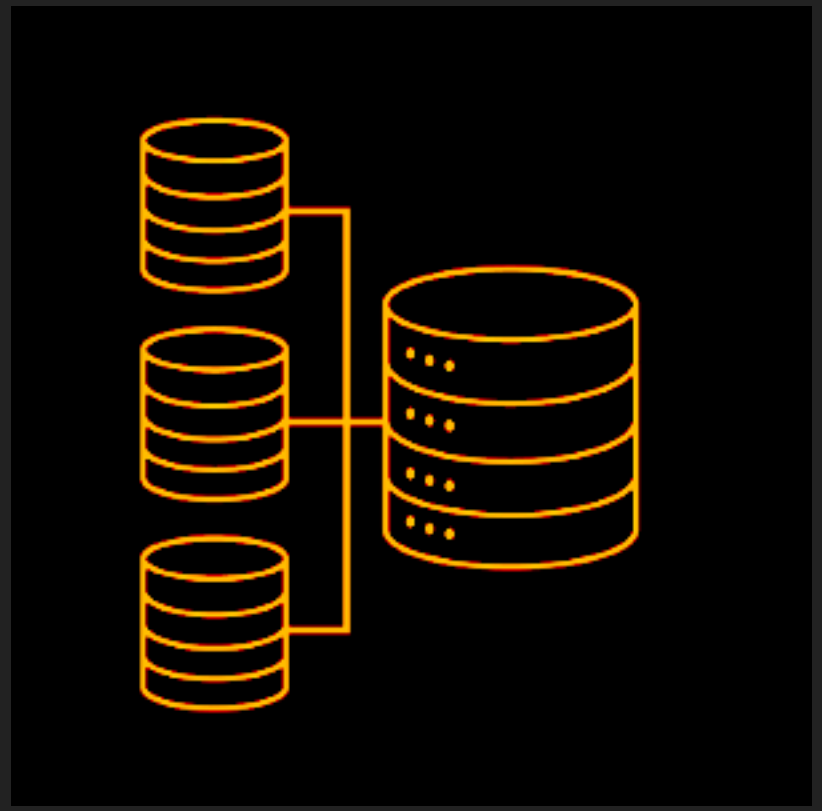
Feature Engineering

- How would you quantify typical behavior in order to predict whether current transaction is an anomaly from typical?
  - Window aggregation features, like average count of transactions at clothing stores in last 30 days
- 4 dimensions: aggregate, entity, filter, window
- Feature space grows exponentially and features are expensive to compute
- 10 000 features implies 10 000 * 1B * 1B computations



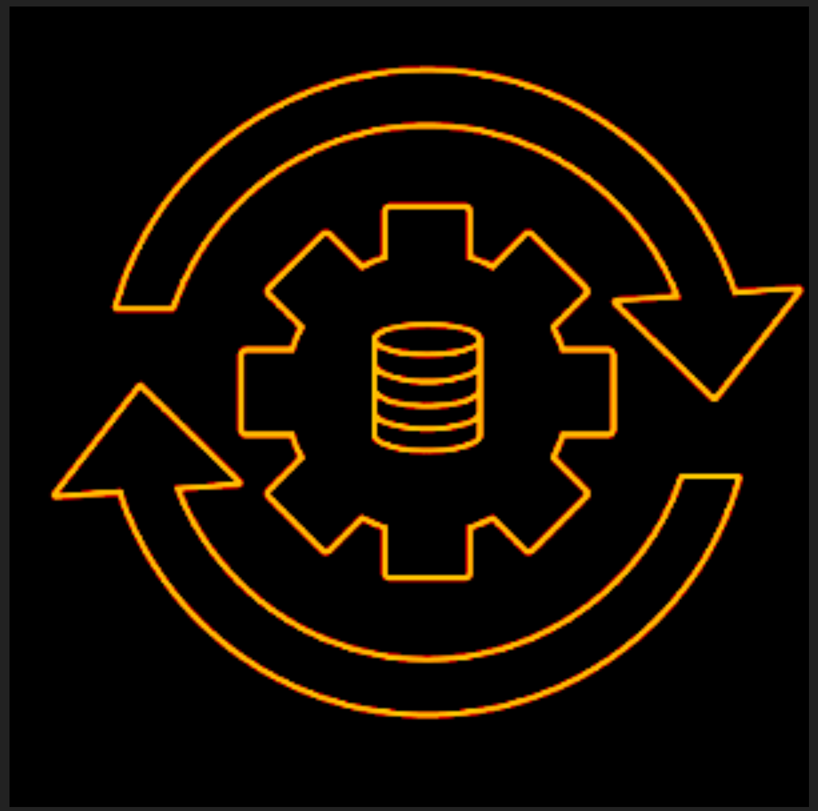


Data Storage, ETL, and Compute



Extract

Retrieve data from various sources

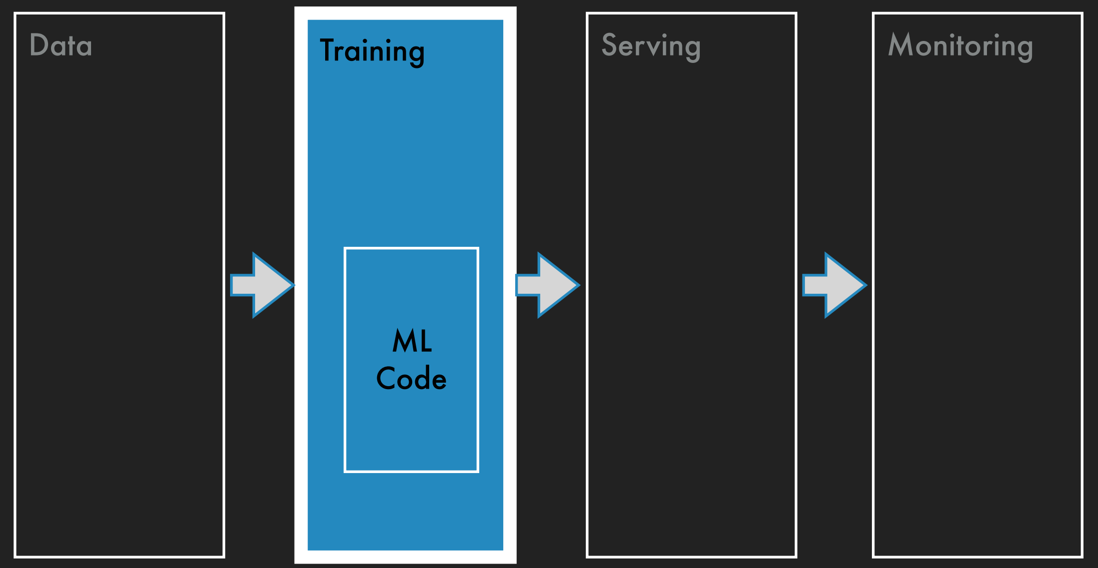


Transform

Process and organize data so it is usable

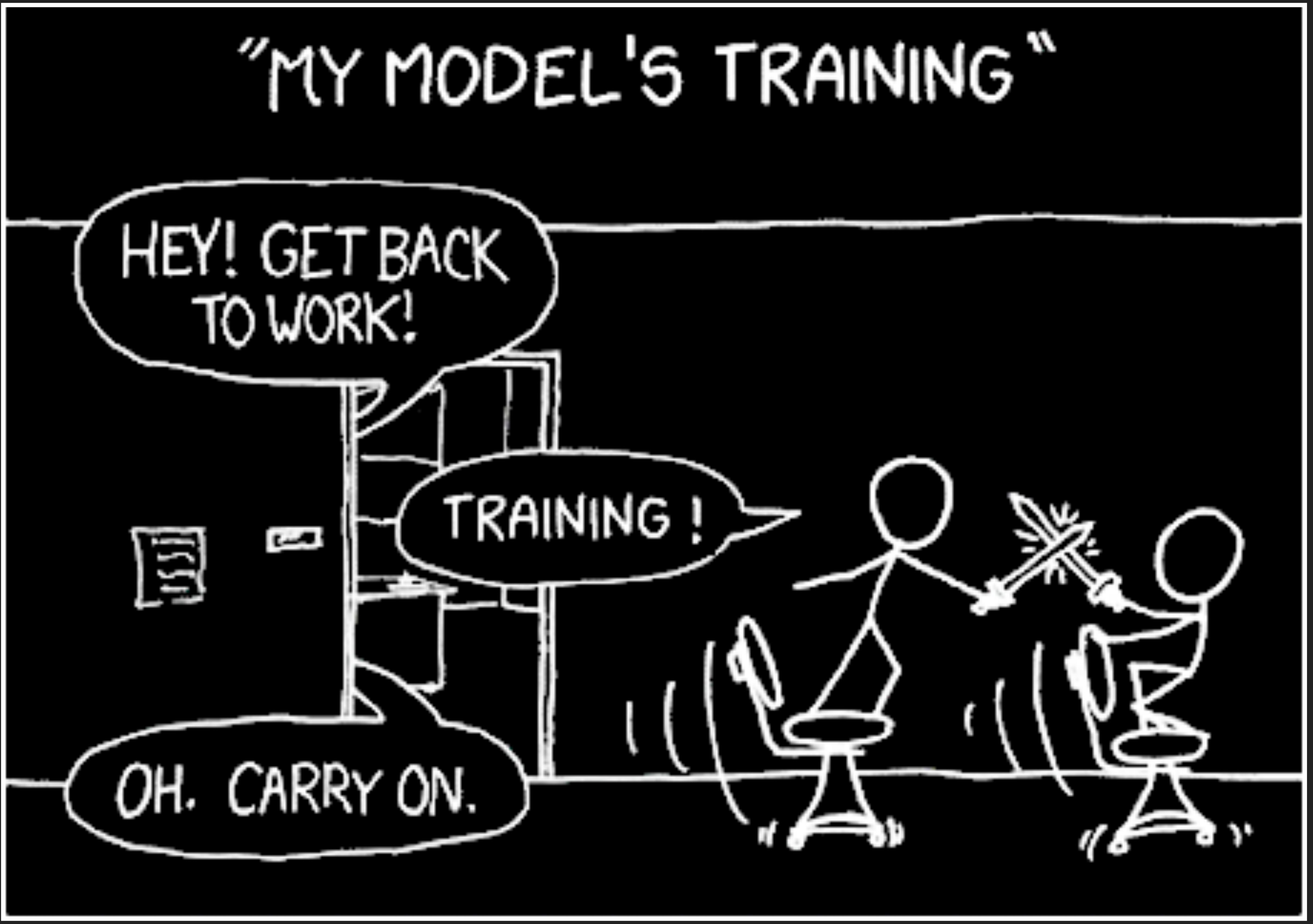
Load

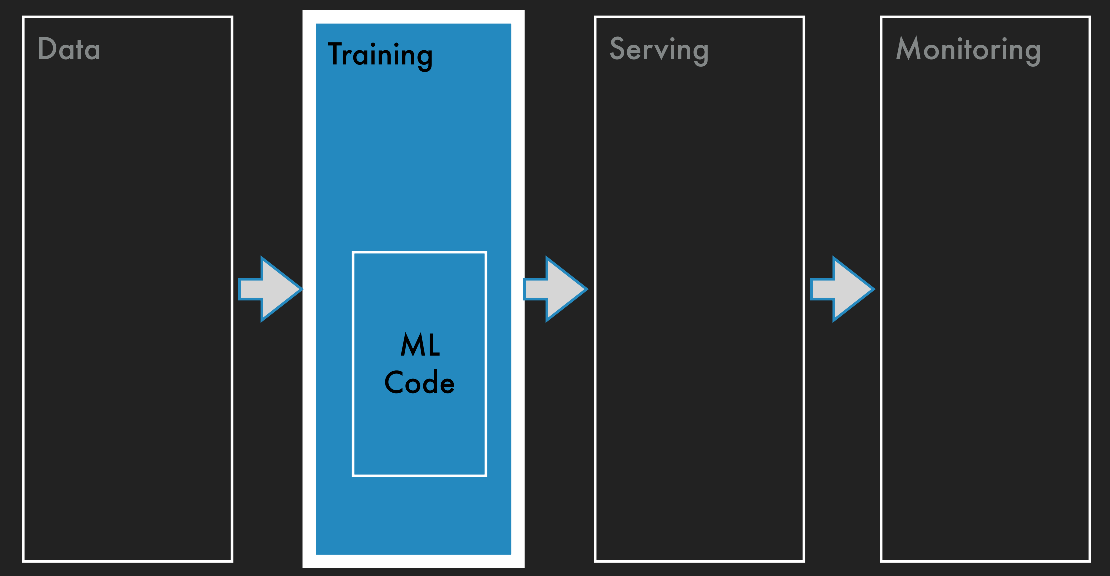
Move transformed data into a repository



Training and Experimentation

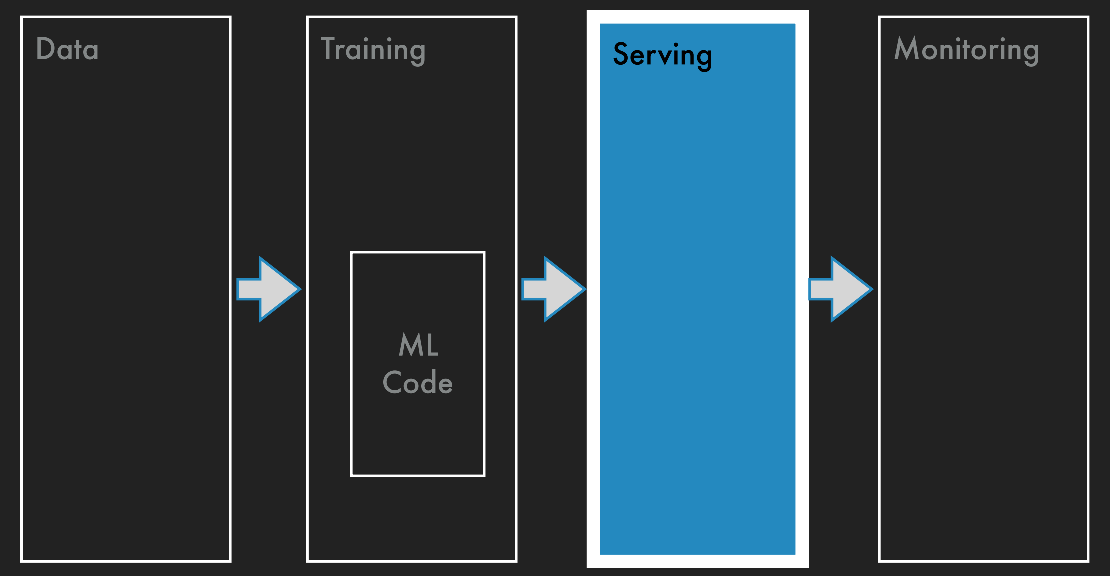
- Metrics
  - Accuracy is not a good metric for imbalanced datasets
  - Precision measures how many predicted positives are in fact positive
  - Recall measures how many of actual positives were we able to capture
  - Precision and recall are conditional on probability threshold, so area under the precision-recall curve is a good scalar measure





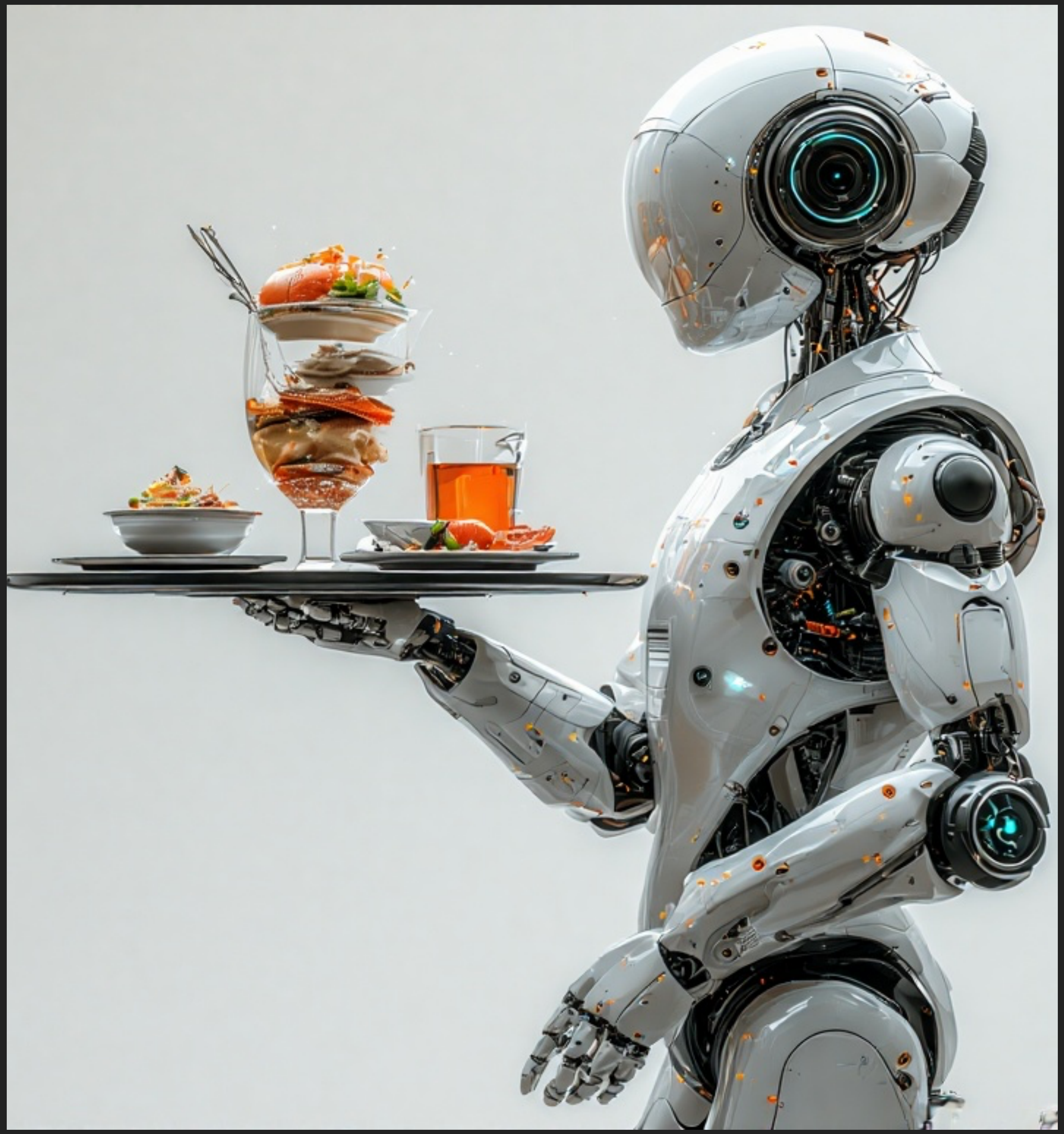
Training and Experimentation

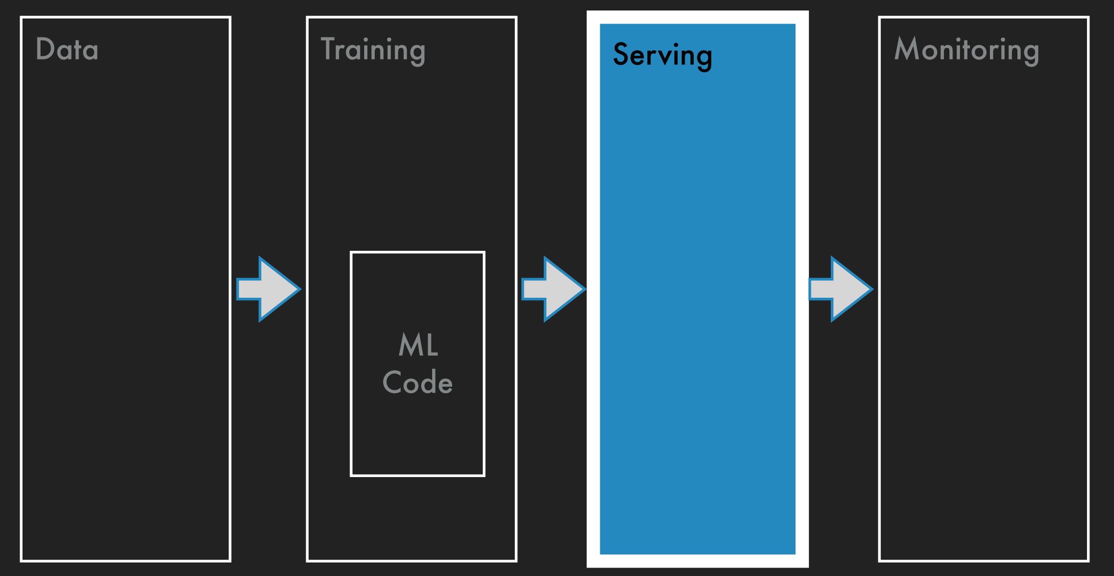
- Metrics
  - Accuracy is not a good metric for imbalanced datasets
  - Precision measures how many predicted positives are in fact positive
  - Recall measures how much of overall fraud were we able to capture
  - Precision and recall are conditional on probability threshold, so area under the precision-recall curve is a good scalar measure
- Training resources
  - Machine size
  - Distributed training
- Experiment tracking
  - Collaborative experimentation
  - Metrics tracking



Model Serving

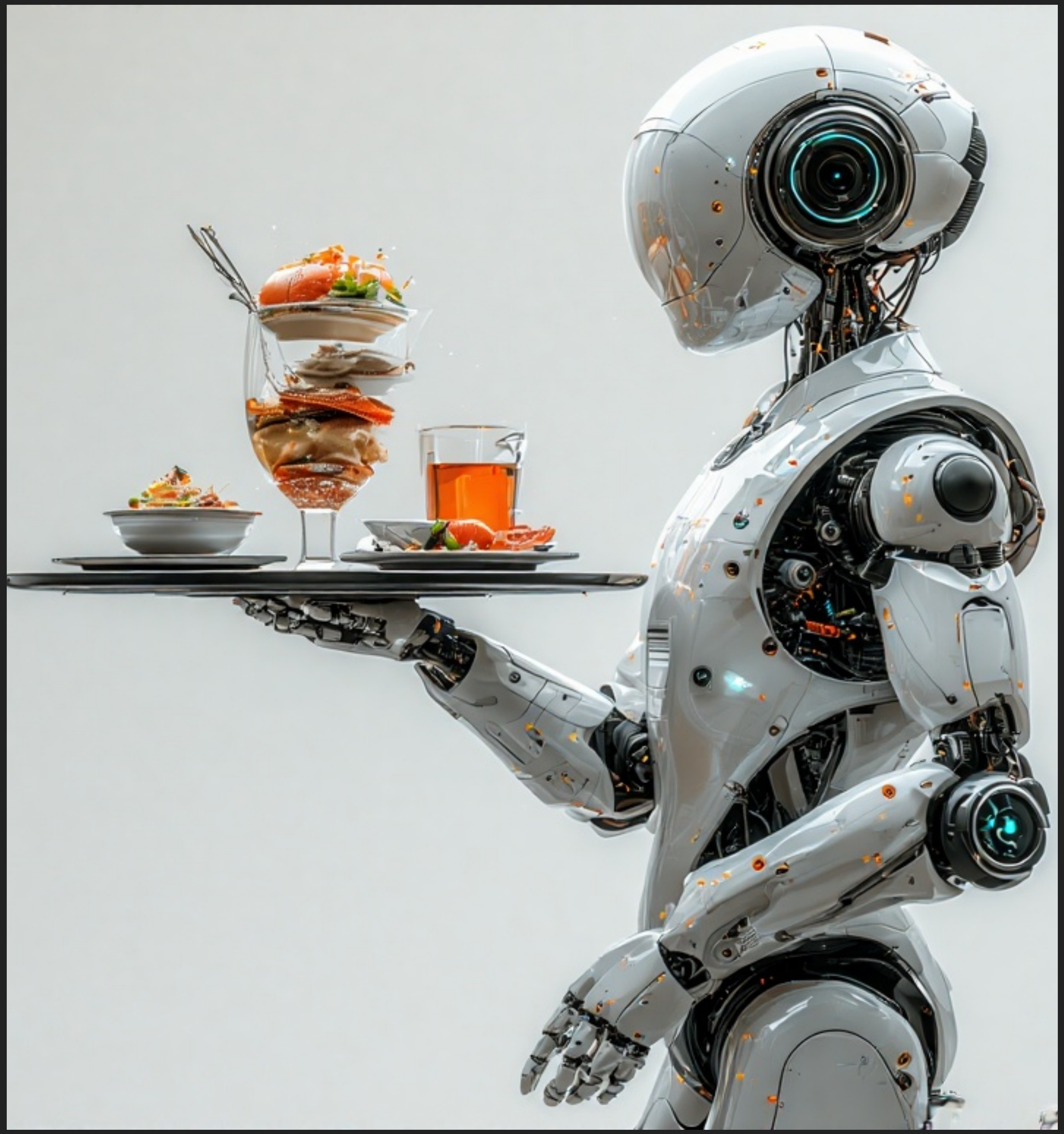
- How does a user access predictions from a trained model?
  - Package the trained model
  - Consider model size, latency, throughput
  - Packaged in memory of the same node as the service using the model
  - Packaged in an endpoint and accessed through an API

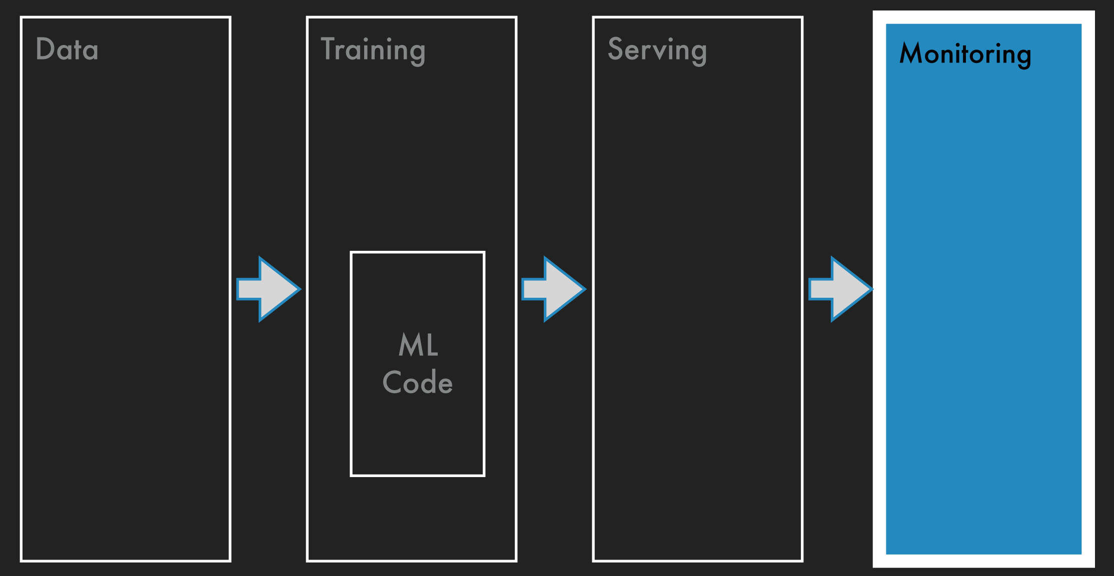




Model Serving

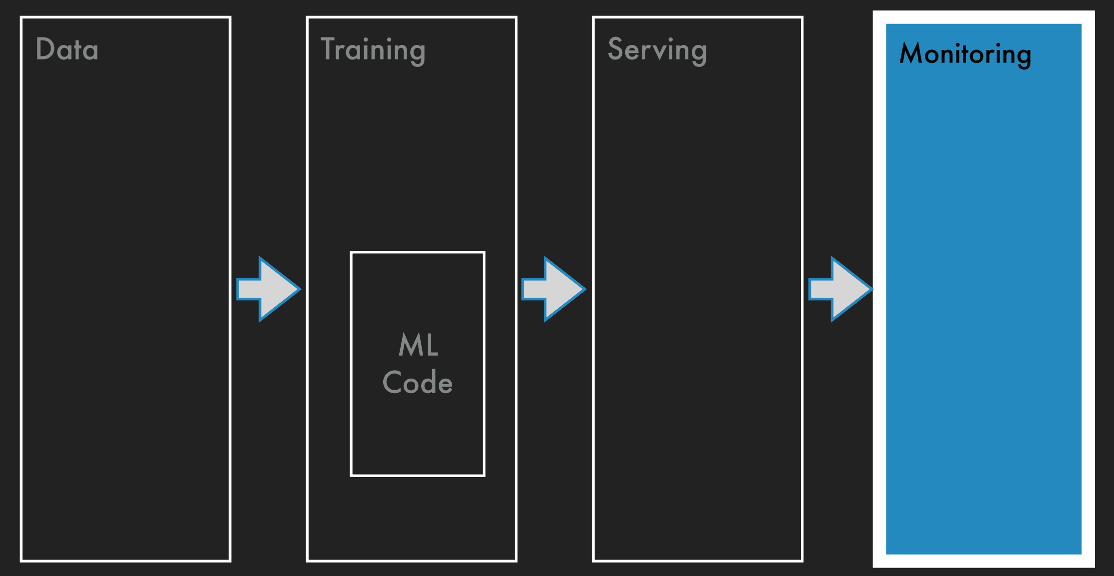
- How does a user access predictions from a trained model?
  - Package the trained model
  - Consider model size, latency, throughput
  - Packaged in memory of the same node as the service using the model
  - Packaged in an endpoint and accessed through an API
- Online and offline features
  - Online features: amount, hour of day
  - Offline features: average amount spent in last 10 days
  - Online transformations: log(amount), distance between buyer and merchant





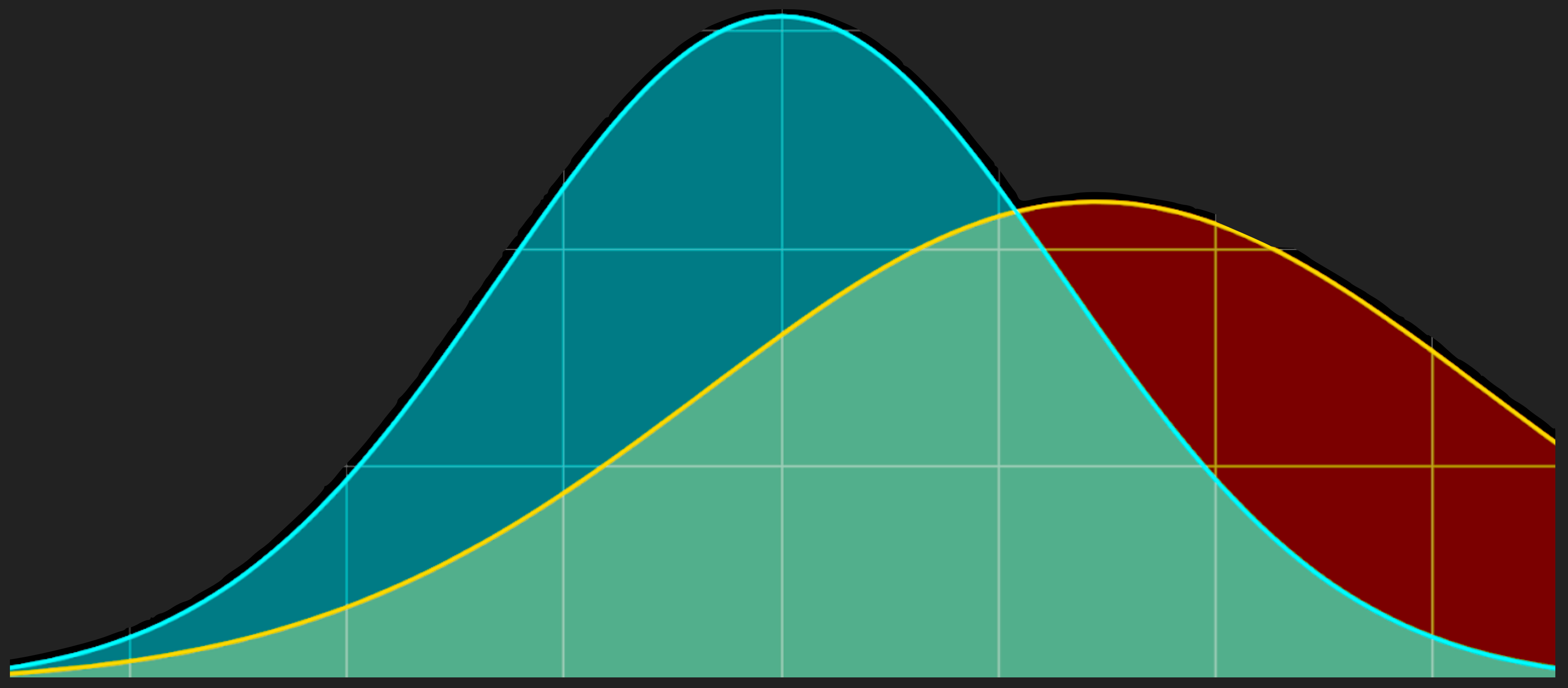
Performance Monitoring

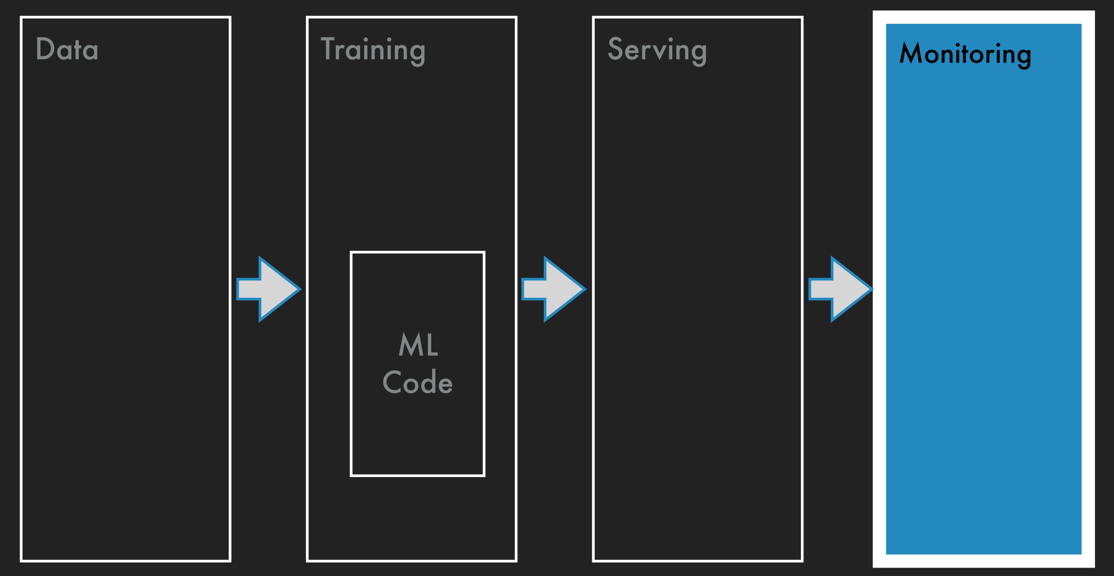
- How will you measure the model's performance on online data?
  - Continuously log latency and throughput of the served model
  - Use chargebacks to measure online performance but chargebacks is a lagging indicator so signals will be delayed
  - Fraud behaviors are continuously evolving, so model performance deteriorates with time



Performance Monitoring

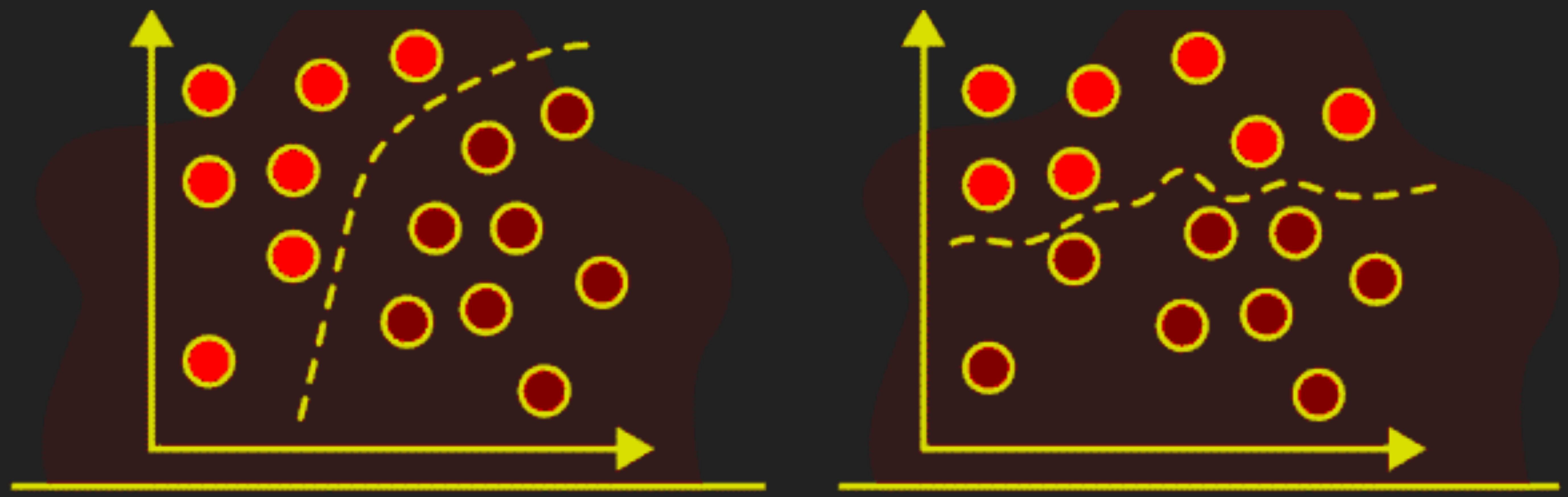
- Data drift
  - Distribution of serving data is different from distribution of training data, e.g., shopping trends changed drastically after COVID lockdowns



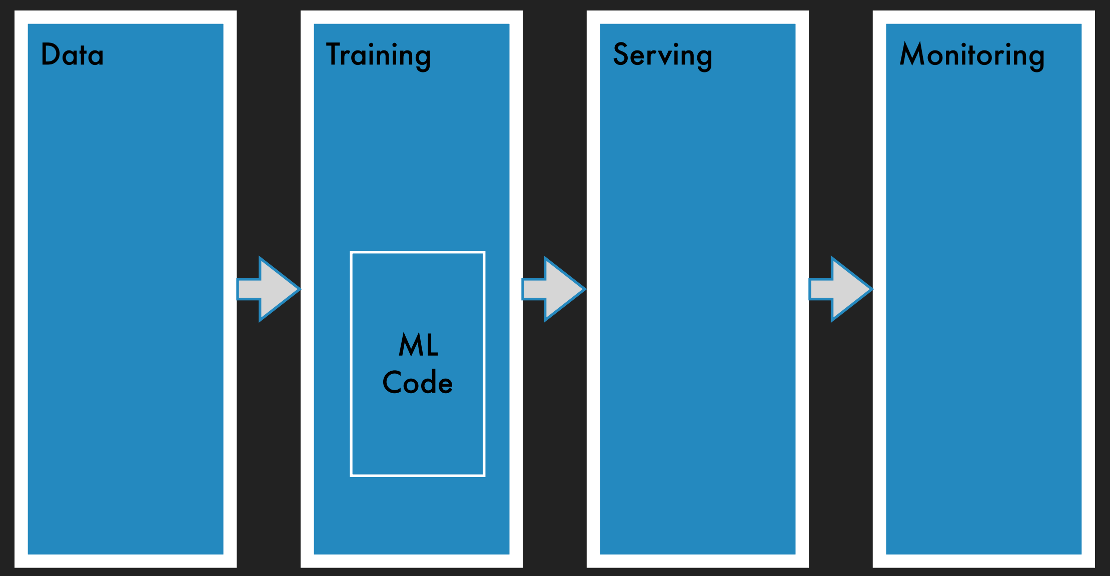


Performance Monitoring

- Concept drift
  - Relationship between features and labels has changed, e.g., number of credit cards per buyer may no longer be a strong fraud predictor due to Apple Pay and Google Pay

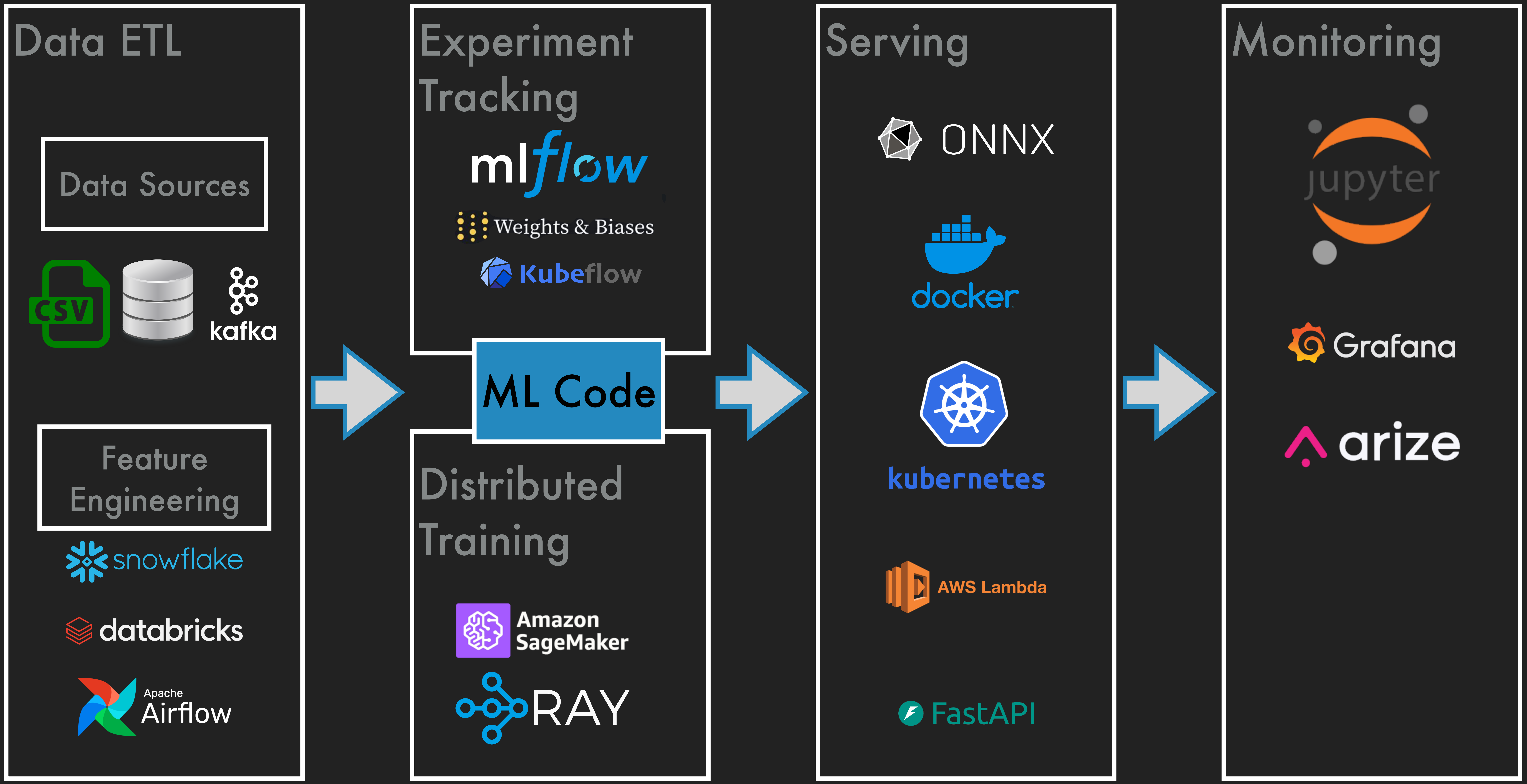


p(y|x) changes



Continuous Integration and Delivery

- Data schema testing
- Data versioning
  - Data definitions can change: boolean columns gets cast to integers
  - Feature definitions can change: last 30 days from time of transaction or calendar days
  - Compare newly trained model with previous model on same evaluation dataset
- Model versioning
  - Multiple models may be simultaneously active: control/treatment, tapered activation
  - Revert to previous model version
- Auto-training models



Data ETL
Data Sources
CSV
kafka
Feature Engineering
snowflake
databricks
Apache Airflow
Experiment Tracking
mlflow
Weights & Biases
Kubeflow
ML Code
Distributed Training
Amazon SageMaker
RAY
Serving
ONNX
docker
kubernetes
AWS Lambda
FastAPI
Monitoring
jupyter
Grafana
arize

Key Takeaways

- ML models are just the tip of the iceberg – often the smallest part of a production ML system
- MLOps is the infrastructure and process around the model that turns prototypes into reliable production systems
- **Plan for production early**: Ask upfront – what are the scale, latency, and throughput requirements of an ML application?

Q/A

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